Roadmap & Acceptance PRD

Implementation phases, acceptance criteria, NFRs, risks and demo narrative

1. Implementation Roadmap

QuickFlow should be delivered incrementally. The roadmap is designed to establish a reliable end-to-end system before adding advanced concurrency, real-time behavior and simulation.

Phase	Primary Outcome
Phase 1 — MVP	Complete order-to-fulfillment path.
Phase 2 — Intelligence	Priority, deadline and hybrid scheduling.
Phase 3 — Concurrency & Resilience	Thread pool, safe reservation, retries and state machine.
Phase 4 — Operations	Real-time dashboard, events and analytics.
Phase 5 — Simulation	Future workload planning and algorithm comparison.

2. Phase 1 — MVP

- Customer login, products, cart and checkout.
- Order creation and persistence.
- C++ engine integration.
- Inventory checking.
- Basic warehouse selection.
- Thread-safe inventory reservation.
- Task generation.
- Basic FIFO scheduling.
- Basic worker execution.
- Admin order visibility.

Exit condition: one order can reliably travel from customer checkout to a completed fulfillment state with persistent status.

3. Phase 2 — Intelligence

- Priority scheduler.
- Deadline scheduler.
- Hybrid scheduler.
- Dynamic deadline-based priority escalation.
- Strategy Pattern for scheduler selection.
- Decision explanation in the admin order view.

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4. Phase 3 — Concurrency & Resilience

- C++ thread pool with fixed worker threads.
- Synchronized task queue and condition variables.
- Mutex/atomic protection for shared inventory.
- Worker availability and assignment.
- Task dependencies.
- Retry queue and retry limits.
- Manual review after repeated failure.
- Order state machine with invalid-transition protection.

5. Phase 4 — Operations Visibility

- Live customer order tracking.
- Live admin task/worker/warehouse updates.
- WebSocket or Socket.IO event propagation.
- Event logging and order timeline.
- Operational analytics.
- Warehouse and worker utilization views.

6. Phase 5 — Simulation & Optimization

- Scenario input for order volume, workers and warehouses.
- Simulation of task workload and scheduling.
- Estimated completion time.
- Expected late orders.
- Worker and warehouse utilization.
- FIFO vs Priority vs Deadline vs Hybrid comparison.

7. Acceptance Criteria

Area	Acceptance Condition
Order	Customer can create and track an order.
Inventory	Concurrent reservations cannot oversell protected stock.
Warehouse	Eligible warehouses are scored and a best candidate is selected.
Tasks	Order is decomposed into dependency-aware tasks.
Scheduler	Multiple strategies can be selected without rewriting orchestration.
Workers	Available workers receive eligible tasks.
Failure	Failed work is retried and can reach manual review.
State	Invalid order transitions are rejected.

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8. Non-Functional Requirements

Requirement	Expectation
Correctness	No unsafe shared-state updates during concurrent inventory operations.
Observability	Important operational actions produce traceable events.
Maintainability	Scheduling and task behavior remain modular through OOP/design patterns.
Scalability	Architecture should allow more orders, workers and warehouses without redesigning the user experience.
Security	Authentication and role-based authorization protect application surfaces.
Deployability	Services can run together using Docker.

9. Key Risks & Mitigations

Risk	Mitigation
Race conditions	Protect critical inventory sections and test concurrent reservations.
Scheduler complexity	Use a common scheduler interface and Strategy Pattern.
Lost tasks after failure	Persist task state and use retry/manual-review flow.
UI becomes too technical	Keep customer and admin experiences intentionally separate.
Simulation produces misleading numbers	Label results as model estimates and make assumptions explicit.

10. Demonstration Narrative

A strong manager demo should start with a real customer order, then follow the same order into the operations console. Show the warehouse decision, the reason for that decision, the inventory reservation, task generation, scheduler ordering and worker assignment. Trigger a status change and show the customer/admin interfaces updating in real time. Finally, open Simulation and demonstrate how changing worker capacity changes projected completion and late-order outcomes.

11. Final Product Positioning

QuickFlow is not positioned as another generic e-commerce application. The e-commerce layer is the entry point. The actual product is the intelligent fulfillment and scheduling engine underneath it.

Its strongest engineering story is the combination of **OOP + DSA + scheduling algorithms + concurrency + safe inventory reservation + failure handling + real-time observability + simulation** in one coherent operational problem.

12. Manager-Level Summary

“QuickFlow is an intelligent order fulfillment and scheduling platform that decides how orders should be fulfilled across a multi-warehouse network. It combines safe concurrent inventory allocation, configurable scheduling, worker management, failure recovery and real-time operational visibility, while providing a simulation layer to evaluate future workload configurations.”

Source note: This PRD consolidates the supplied QuickFlow handwritten project notes and the collected text specification. Proposed techniques remain proposals until implemented and validated.